WHY WILDFIRE?

Next-day wildfire spread prediction from satellite data · Phase 1 · TensorTitans, Section C

Day t+1

Real-world impact

Evacuations and fire crews are planned a day ahead. Responders need tomorrow's fire map, not today's.

13 layers

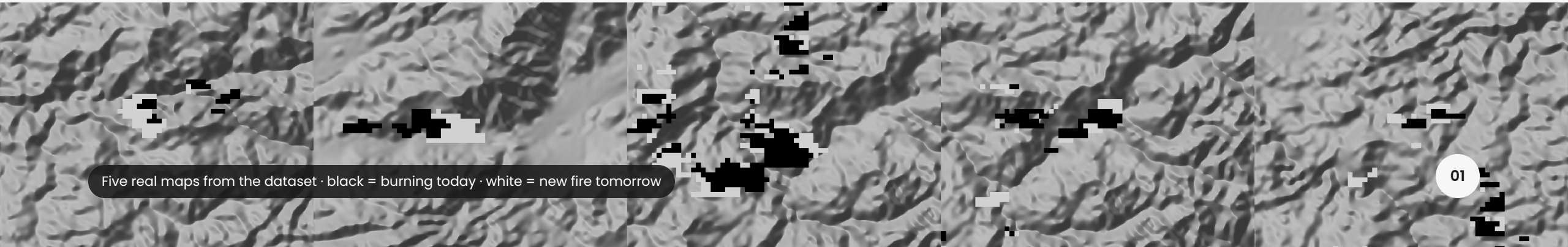
Real, public data

Weather, terrain, fuel and fire for every 1 km square. Huot et al. (2022), IEEE TGRS, US fires 2012–2020.

0.08–1.09%

A hard ML problem

Share of pixels on fire the next day. So rare that accuracy is misleading and neighbours look alike.



Five real maps from the dataset · black = burning today · white = new fire tomorrow

THE DATA

The dataset (Huot et al., 2022) ships as 19 TFRecord files. Each file holds many 64 × 64 km maps, and every 1 km square becomes one table row.

19 FILES



- **15** train · 3,047 MB
- **2** validation · 382 MB
- **2** test · 344 MB

Whole files go to one split, so a fire never appears in two splits.

HOW WE GOT THE ROWS

STEP	TRAIN	VALIDATION	TEST
1 Maps read from the files	30	10	10
2 Squares (maps × 4,096)	122,880	40,960	40,960
3 Remove unobserved squares (-1)	-5,729	-11,168	-8,093
4 Maps with fire: all fire + 10 no-fire per fire	682 10 maps · 62 + 620	429 1 map · 39 + 390	1,747 5 maps · 157 + 1,590
5 Maps without fire: every square kept	76,562 20 maps	25,696 8 maps	12,682 5 maps
Rows used (step 4 + step 5)	**77,244**	**26,125**	**14,429**
of which on fire next day	62	39	157

Sampling only thins maps that contain fire, so fire-free maps supply most rows. One validation map was fully unobserved (9 maps remain), and all 39 validation fires come from a single map.

THE COLUMNS

What is r (train / test)?

The correlation between a column and next-day fire, from -1 to +1, where 0 means no linear link. We compute it twice: on the train rows and on the test rows. A column only counts as useful if r is clearly above 0 in both, like **PrevFireMask .12 / .31**.

How are the groups made?

By type of information, added one layer at a time, so the ablation shows what each type adds:

- **A** Weather on day t — 6 features
- **B** A + fire today — 7 features
- **C** B + land, fuel, people — 12 features
- **D** C + engineered features — 18 features

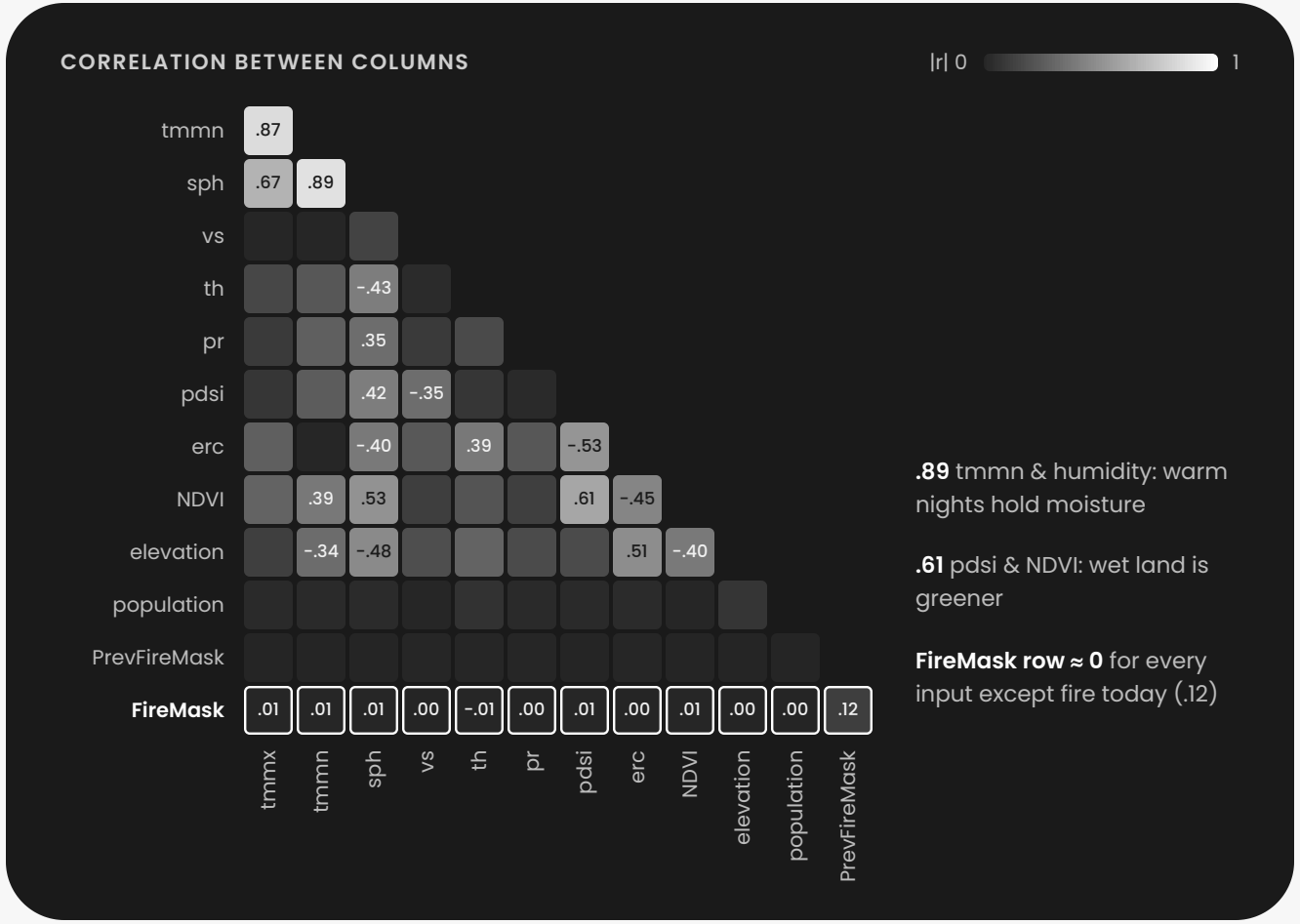
COLUMN	FULL NAME	UNIT	USED FOR	GROUP	R TRAIN / TEST
elevation	Elevation	m	Terrain height; fire runs uphill	C	.00 / .28
th	Wind direction	deg	Which way fire is pushed	A	-.01 / .10
vs	Wind speed	m/s	How hard fire is pushed	A	.00 / -.10
tmmn	Minimum temperature	K	How warm the night stays	A	.01 / -.02
tmmx	Maximum temperature	K	Daytime heat that dries fuel	A	.01 / .07
sph	Specific humidity	kg/kg	Moisture in the air	A	.01 / -.07
pr	Precipitation	mm/day	Rain dampens fuel	A	.00 / -.07
pdsi	Palmer Drought Severity Index	index	Long-term dryness	C	.01 / .07
NDVI	Normalized Difference Vegetation Index	index	Amount of green fuel	C	.01 / -.01
population	Population density	per km²	Human ignitions	C	.00 / -.01
erc	Energy Release Component	index	Heat dry fuel would release	C	.00 / .20
PrevFireMask	**Fire mask, today (day t)**	**0 / 1**	**Is this square burning now**	**B**	**.12 / .31**
FireMask	Fire mask, tomorrow (day t+1)	0 / 1	The label we predict	—	—

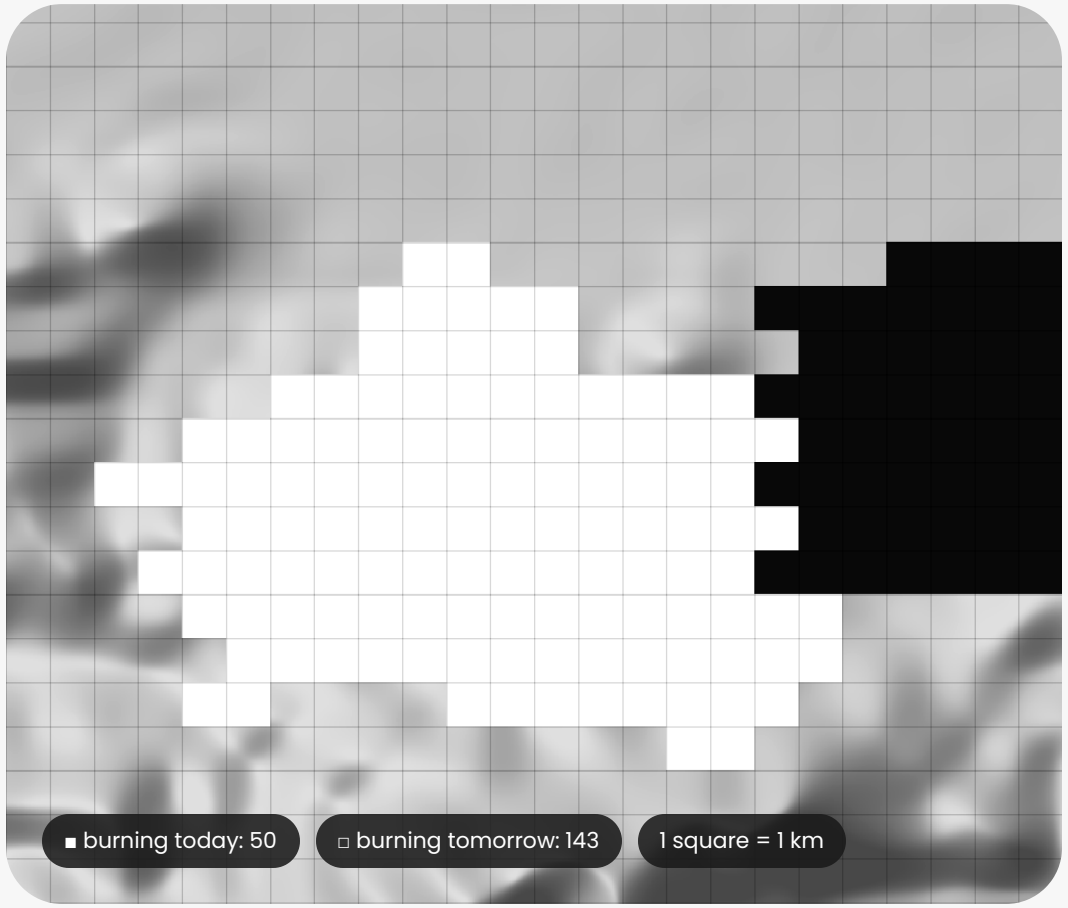
EDA

Exploratory data analysis, train set

AVERAGE VALUE	NO FIRE NEXT DAY	FIRE NEXT DAY
Max temperature (K)	296.7	299.3
Wind speed (m/s)	3.64	3.67
Humidity (kg/kg)	0.007	0.008
Fuel dryness, ERC	38.0	36.0
Burning today	**0.13%**	**16.1%**

Squares that burn tomorrow are **120×** more likely to be burning today, while every weather gap is small. 0 missing values. Based on 62 fire squares.





FEATURE ENGINEERING

Real test map #5 (left): new fire appears right next to existing fire. We turned that into features.

temp_range	tmmx - tmmn: big swing = dry air
wind_u, wind_v	wind as x/y, so 359° and 1° are both north
drought_fuel_index	ERC × drought, combined
fire_neighbor_3x3 / 5x5	**burning squares around each square today**

BURNING NEIGHBOURS, AVERAGE



No fire next day (top) vs fire next day (bottom): **90×** higher. **Bug fixed:** counts were taken after rows were sampled away; now from the full map. r with label: train .065 → .154, test .374.

01 · FLOOR

Persistence

Tomorrow = today. Does learning beat a rule?

PR-AUC	ROC-AUC	
0.0015	0.498	coin flip

02 · LINEAR

Logistic Regr.

Explainable weights. Is a straight line enough?

PR-AUC	ROC-AUC	
0.0008	0.141	backwards

03 · MAIN MODEL

LightGBM

Boosted trees. Do feature combinations help?

PR-AUC	ROC-AUC	
0.0016	0.518	best → test

04 · CROSS-CHECK

XGBoost

Second tree library. Or one library's quirk?

PR-AUC	ROC-AUC	
0.0009	0.199	backwards

THE MODELS

Validation set, 39 fire squares. Random guessing scores PR-AUC 0.0015 and ROC-AUC 0.500. Each model answers one question, from a no-learning rule to boosted trees.

Why LightGBM won

The only model above 0.5 ROC-AUC. Trees combine clues (hot *and* a neighbour burning); **scale_pos_weight = 8** weights the rare fires.

But only just

0.0016 vs 0.0015 for random guessing, so it was chosen for the test, not proven.

FROZEN TEST SET · LIGHTGBM, GROUP D

	ACTUAL FIRE	ACTUAL NO FIRE
PREDICTED FIRE	0 fires caught	96 false alarms
PREDICTED NO FIRE	157 fires missed	14,176 correct

MODEL ACCURACY
98.25%

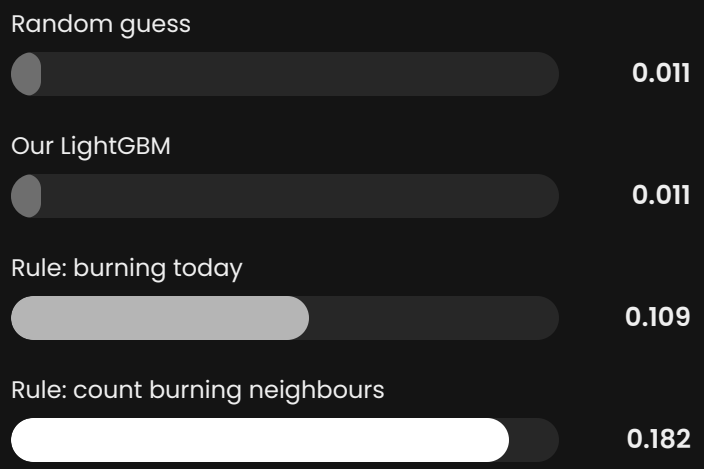
ALWAYS "NO FIRE"
98.91%

THE RESULTS

TEST METRIC	LIGHTGBM	RANDOM GUESS
PR-AUC	0.0109	0.0109
ROC-AUC	0.495	0.500
Recall at 5% false alarms	0.00	0.05

Why not accuracy: 99% of squares are "no fire", so never predicting fire scores 98.91%, higher than our model. PR-AUC and ROC-AUC can't be fooled that way. Threshold tuned on validation only; test used once.

TEST PR-AUC · HIGHER IS BETTER



A one-line rule using our neighbour feature is **17×** better than random (ROC-AUC 0.632 vs the model's 0.495).

KEY INSIGHTS

The signal is in the data

Counting burning neighbours alone reaches PR-AUC 0.182 on test. Our model didn't learn it yet.

Too few fires to learn from

Only 62 fire rows in train. The final model gives only 58 different output values for 14,429 test squares.

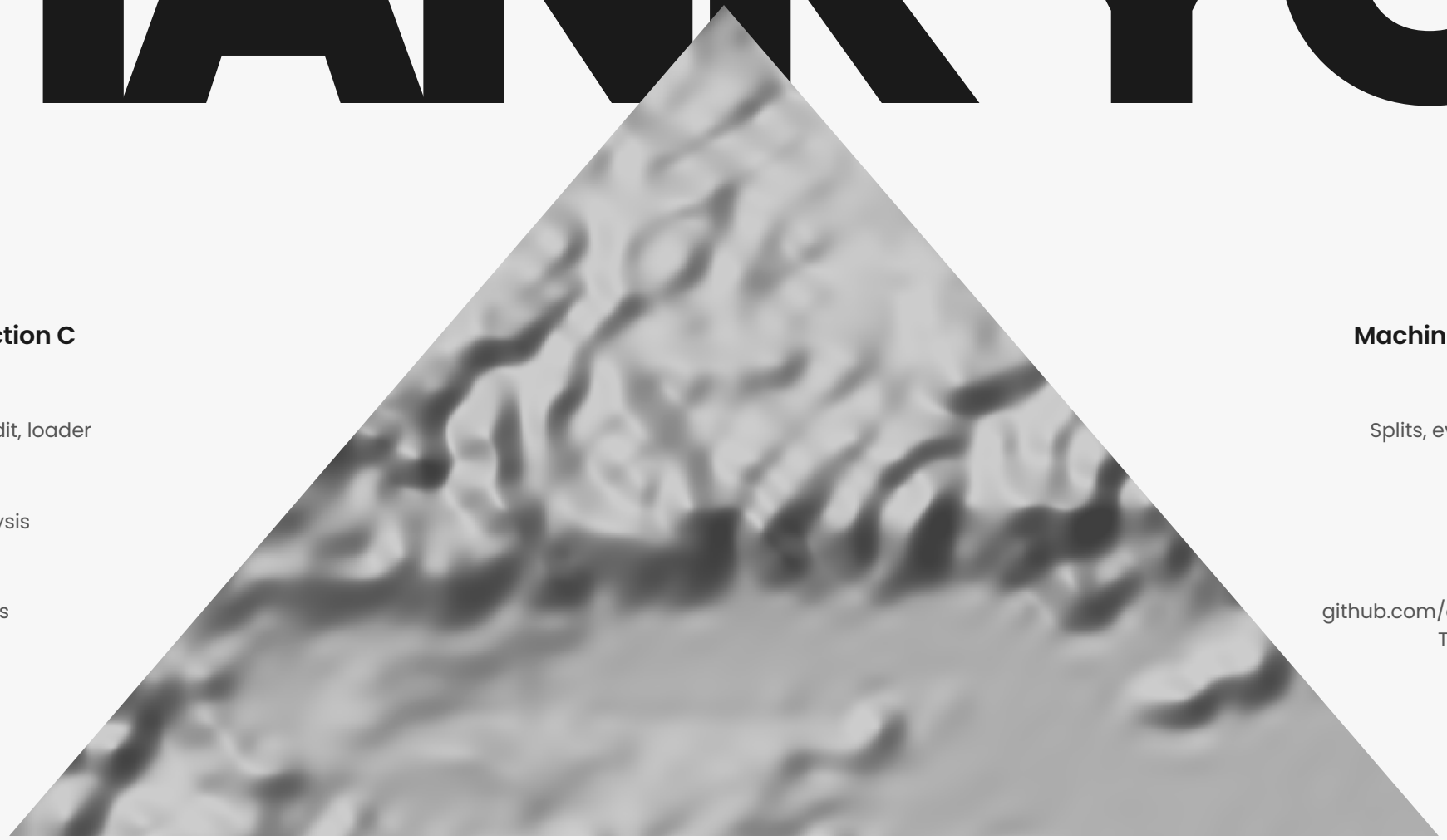
Fire today is the real driver

All 11 weather and land columns have $|r| \leq 0.015$ with next-day fire on train.

Next steps

Hundreds of maps per split, early stopping, threshold tuned on the same model, then Phase 2: a U-Net on full maps.

THANK YOU



Presented by

TensorTitans · Section C

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Literature, dataset audit, loader

Riya
Exploratory data analysis

Krish
Features, error analysis

Phase 1

Machine Learning Project

Nitanshu
Splits, evaluation, comparison

Tejas
Models, evaluation

Code
github.com/connectwithvanshika/TensorTitans_SectionC